Beyond Spatial Accuracy: Diagnosing Persistent Orientation Failures in Vision–Language Models

Anonymous WACV Datasets Track submission

Paper ID *****

Abstract

Aggregate spatial accuracy can hide sharply different capabilities within vision–language models (VLMs). We study object-relative orientation as a diagnostic case. On Visual Spatial Reasoning (VSR), moving from a compact 2.2B model to a larger 7B VLM substantially improves overall, depth, horizontal, and containment accuracy, but orientation changes only modestly. Except for structured prompting, which degrades performance, trained conditions on both models remain in a narrow 62–66% orientation band. We test structured prompting, general and targeted LoRA, hard negatives, visual/projector adaptation, frozen and object-grounded representation probes, and object-centric decomposition; none of the tested configurations reliably closes the gap, and a single-annotator clean-label sensitivity analysis shows that the pattern persists after excluding examples flagged as ambiguous.

Accuracy also fails to capture relational coherence. LM-side adaptation significantly improves consistency over zero-shot, while hard-negative training raises facing/facing-away consistency from 66.0% to 77.7% with facing accuracy unchanged at 68.9%; the additional pooled consistency gain over LM-only is not significant. Cross-dataset evaluation on SITE shows that VSR-trained adaptation does not transfer cleanly: it is neutral overall and significantly harms official spatial-relationship reasoning (75.1%→71.6%, $p = 0.004$). A pre-specified orientation-vocabulary slice, frozen before SITE evaluation, is analyzed as exploratory evidence rather than as direct replication of the VSR construct. The results point to a persistent orientation bottleneck in which orientation is not robustly decodable by the tested frozen-feature readouts and relational decisions are not always propagated coherently, while also showing that task-specific spatial adaptation can transfer poorly.

1. Introduction

Spatial reasoning is central to multimodal systems that must interact with physical environments. A useful VLM should not only recognize objects, but determine whether one object is in front of another, whether it is facing another, whether two objects are parallel, and whether equivalent or complementary formulations yield logically compatible answers. These abilities matter for robotics, embodied agents, navigation, and scene understanding.

Existing work already shows that spatial relations are difficult for vision–language systems. VSR was designed as a controlled natural-image probe and reported particularly poor performance on object-orientation relations in earlier VLMs [7]. More recent controlled benchmarks likewise find that strong VLMs can fail on basic left/right and other grounded relations [6, 9], while broader evaluations continue to identify spatial orientation as a difficult factor [14]. Our question is therefore not whether spatial reasoning can fail, but whether the orientation weakness persists in modern generative VLMs of different sizes and after targeted intervention, and what the failure signature reveals about representation, reasoning, and transfer.

In our VSR experiments, the distinction between relation families is striking. Moving from SmolVLM2-2.2B-Instruct to the larger Qwen2-VL-7B-Instruct increases zero-shot overall accuracy from 74.0% to 80.9%. Horizontal reasoning rises from 70.2% to 84.8%, depth from 68.9% to 75.2%, and containment from 83.4% to 89.3%. Orientation, by contrast, moves only from 62.8% to 63.5%. We then evaluate interventions spanning prompting, language-side adaptation, targeted data construction, visual-side adaptation, representation probing, and explicit object-centric decomposition. Structured prompting significantly hurts performance; general and targeted LoRA improve aggregate accuracy but do not remove the orientation ceiling; hard negatives improve targeted relational coherence without a significant overall orientation-accuracy gain; and the tested visual-side LoRA configurations do not outperform the language-side control.

As a sensitivity analysis, removing examples flagged by a single-annotator audit raises orientation accuracy (65.7%→78.5% for 7B General LoRA, 66.4%→76.6% for hard-negative LoRA on the strictest 107-example subset) but does not eliminate error (2B structured shows no gain). Because the removed examples were drawn from persistent failures, part of this gain is expected by construction.

A second failure mode emerges from logical consistency. The zero-shot 7B model is consistent on only 36.9% of complementary facing/facing-away pairs. LM-side LoRA substantially improves coherence, and hard-negative training raises facing consistency to 77.7% while facing accuracy remains unchanged at 68.9%. The extra hard-negative gain is not significant in the pooled strict-family comparison, yet accuracy stays unchanged while consistency moves: the two metrics capture different behavior.

Finally, we test cross-dataset transfer. The 7B model is strong on SITE's official spatial-relationship-reasoning category (75.1% raw; 59.2% chance-adjusted accuracy), yet the VSR-trained adapter does not generalize cleanly: it is statistically neutral overall (54.2%→53.8%, $p = 0.66$) and significantly degrades the official category (75.1%→71.6%, $p = 0.004$). A pre-specified, non-official orientation-vocabulary slice is much weaker in the aggregate (48.3% raw; 24.0% chance-adjusted) and shows no positive transfer, but—as we detail below—its composition limits what it can establish, and we therefore do not treat SITE as independent confirmation of the VSR orientation construct.

Our contributions are fourfold:

- Fine-grained diagnosis across two differently sized VLMs. Across ten VSR conditions and two differently sized models, several spatial families improve substantially while object-relative orientation remains unusually resistant.
- Robustness and mechanism analysis. A single-annotator clean-label sensitivity analysis, frozen/object-grounded probes, visual-side adaptation, and object-centric decomposition constrain purely annotation-based or single-component explanations.
- VSR-specific accuracy–coherence separation. Complementary-relation tests on VSR's strict relation pairs show that, for the tested conditions, relational consistency can change substantially while task accuracy remains fixed.
- Cross-dataset transfer analysis. SITE shows strong official spatial-relationship performance, while VSR-trained LoRA transfers poorly and significantly harms the official SITE category; the orientation-vocabulary slice is analyzed as exploratory evidence.

2. Related Work

Spatial reasoning benchmarks. VSR introduced a controlled natural-image benchmark of more than 10k image–text pairs spanning 66 spatial relations and explicitly highlighted orientation relations such as facing and parallel as difficult for the VLMs studied at the time [7]. Our work should therefore be read as a modern diagnostic extension rather than the first observation of an orientation weakness. What'sUp isolates basic spatial relations by holding object identities fixed while varying configurations, and found large gaps between humans and tested vision–language models [6]. GSR-Bench extends controlled grounded spatial evaluation across a larger set of multimodal models and model scales [9]. SITE broadens evaluation to single-image, multi-image, and video inputs and organizes tasks using cognitive-science dimensions including spatial visualization/orientation, intrinsic/extrinsic relations, and static/dynamic settings; its authors report spatial orientation as a persistent challenge [14]. Recent broad evaluations such as SpatialScore similarly argue that spatial intelligence remains fragmented across tasks and modalities [15], and Mind the Gap isolates spatial reasoning from detection and semantic comprehension in contemporary VLMs [10].

Diagnosing spatial failures. Recent work has begun to analyze spatial failures mechanistically rather than only benchmarking them. On VSR itself, AdaptVis traces attention distributions and finds that successful spatial reasoning correlates with attention aligning to actual object locations [2]. CREG characterizes the directional organization of attribution evidence in Qwen2-VL on spatial-relation benchmarks, showing that standard heatmaps often reflect image layout rather than the queried relation [11]. When More Is Less analyzes when injecting spatial cues, commonsense knowledge, or chain-of-thought instructions helps or hurts VSR, with effects varying by model and setup [1]. Probing Visual Concepts measures linear decodability of visual concepts, including orientation-relevant ones, in lightweight driving VLMs and identifies information-flow bottlenecks [12]. VSP decomposes visual spatial planning into perception and reasoning subtasks [16], and MM-Spatial targets 3D spatial understanding with geometric inputs [3].

Prior work has thus already established spatial and orientation weaknesses in VLMs and has begun to analyze those failures through prompting, attention, representation probing, and consistency-based training. Our contribution is therefore not the first identification of orientation difficulty or spatial inconsistency. Instead, we revisit the known VSR orientation weakness in modern generative VLMs and subject the same relation family to a controlled diagnostic ladder spanning adaptation, frozen and object-grounded readouts, visual-side interventions, complementary-relation consistency, label sensitivity, and cross-dataset transfer.

Adaptation and spatial specialization. We use LoRA, which adapts frozen models through trainable low-rank updates [5], to compare general, targeted, hard-negative, projector, and vision-plus-projector interventions. This design lets us ask whether benchmark gains reflect general spatial competence or specialization. The external SITE transfer experiment is central here: VSR adaptation is neutral overall on SITE, does not improve the orientation subset, and significantly degrades SITE's official spatial-relationship category. The result complements benchmark work by explicitly measuring negative as well as positive cross-dataset transfer.

Representation probing and logical consistency. Diagnostic probes can reveal whether information is accessible from frozen representations, but probe performance must be interpreted cautiously because probe capacity can itself learn the target task [4]. We therefore use linear and nonlinear probes, compare against majority baselines, add object-grounded features, and avoid claiming that weak decodability proves information is absent. Separately, consistency across transformed or complementary statements has prior literature, including consistency-oriented training objectives such as SAGE's geometric and linguistic duality operations [8]. Our consistency analysis is a VSR-specific diagnostic rather than a new consistency framework: it uses strict complementary relation pairs with observed verdict-pattern and base-rate analysis, and it shows that instance-level accuracy and complementary-relation coherence can diverge for the tested conditions.

3. Experimental Setup

3.1. Benchmarks

VSR. We use VSR [7] as the primary diagnostic benchmark. The random-split test set contains 2,195 examples. The benchmark defines 64 relation labels; 61 appear in the evaluated test split. The orientation family contains facing, facing away from, parallel to, and perpendicular to, totaling 137 test examples. All main VSR results use the same held-out test set and deterministic True/False parsing.

SITE. We use SITE [14] for cross-dataset transfer evaluation only and do not train on SITE. The image evaluation contains 2,591 examples: 1,368 single-image and 1,223 multi-image examples. Our primary SITE subset is the official spatial-relationship-reasoning category ($n = 993$). We additionally report a transparently defined, non-official keyword-derived orientation subset ($n = 2{,}024$), analyzed as exploratory evidence rather than as direct replication of the VSR construct. SITE results use the benchmark's native multiple-choice format and chance-adjusted accuracy (CAA) in addition to raw accuracy.

3.2. Models and Conditions

Our compact model is SmolVLM2-2.2B-Instruct, and our larger model is Qwen2-VL-7B-Instruct [13]. We evaluate ten canonical VSR conditions: 2B zero-shot, 2B structured prompting, 2B General LoRA, 2B Targeted LoRA, 7B zero-shot, 7B General LoRA, 7B Targeted LoRA, 7B Hard-Neg LoRA, 7B Projector LoRA, and 7B Vision+Projector LoRA. LoRA [5] is used for all parameter-efficient adaptation conditions.

General LoRA uses a proportional sample across relation families. Targeted LoRA allocates most training examples to weak orientation/depth/horizontal families with rehearsal from the remaining relations. Hard-negative LoRA introduces audited complementary orientation examples, primarily facing/facing-away inversions. Vision-side conditions adapt either the multimodal projector or upper vision blocks together with the projector. The frozen experimental snapshot records the final result set.

3.3. Metrics and Statistical Tests

For VSR we report accuracy overall and by spatial family. For SITE we report raw accuracy and CAA. Paired condition comparisons use McNemar tests when applicable. Clean-label orientation analysis reports the full orientation set and progressively stricter audited subsets. Logical consistency is evaluated by constructing complementary statements for strict relation pairs (left/right, front/behind, facing/facing-away) and testing whether the model assigns opposite verdicts as required. Parallel/perpendicular is treated separately as a soft complement because both relations can be false for oblique objects.

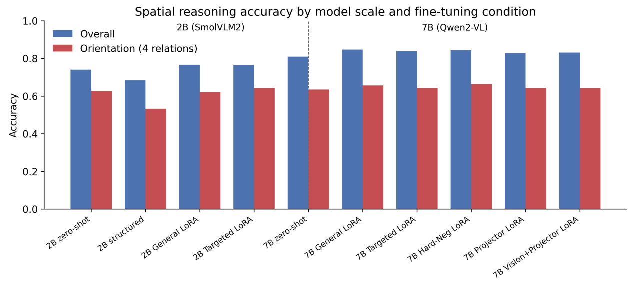

Figure 1. Overall versus orientation performance across the two VLMs and adaptation conditions.

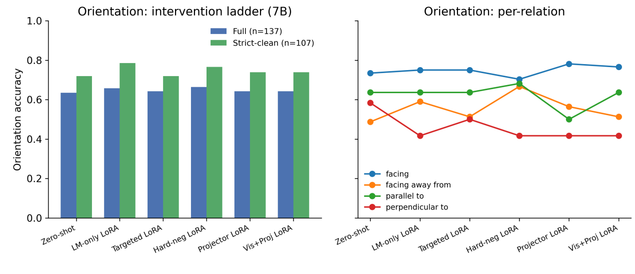

Figure 2. Orientation performance across 7B interventions, including clean-label analysis and per-relation behavior.

4. Main Results

4.1. Orientation Remains Weak Across Two Differently Sized VLMs

Table 1 shows the central pattern. Moving from the 2B model to the 7B model lifts zero-shot overall accuracy by 6.9 pp, horizontal by 14.6 pp, depth by 6.3 pp, and containment by 5.9 pp, but orientation by only 0.7 pp. General 7B LoRA further raises overall accuracy to 84.7%, yet orientation reaches only 65.7%. Across trained conditions other than structured prompting, orientation remains in a narrow 62–66% band.

Structured prompting is a negative result: overall accuracy falls from 74.0% to 68.3% at 2B, while orientation falls from 62.8% to 53.3%. A paired exact McNemar test on the full test set confirms that the structured prompt is worse than the minimal baseline ($p \approx 2.2 \times 10^{-7}$). Consistent with recent analyses of information injection on VSR, where additional spatial cues or reasoning instructions can help or hurt depending on the model and setup [1], this result is specific to the tested 2B condition and prompt; it does not establish that prompting generally fails.

4.2. Clean-Label Sensitivity Analysis

A single-annotator exploratory audit identified ambiguous or questionable orientation examples, so we repeat evaluation on progressively stricter subsets as a sensitivity analysis (full table in the supplementary). The best full-set orientation result is 66.4%; on the strictest 107-example subset, the best result rises to 78.5% for 7B General LoRA. Because auditing changes the evaluated sample, these values should not be read as a matched comparison against other relation families. The robust conclusion is narrower: removing flagged examples raises orientation scores but does not eliminate error across conditions; the audit is exploratory qualitative evidence, not a ruling on label noise.

5. Why Does Orientation Remain Difficult?

5.1. Frozen Representation Probes

We probe frozen Qwen2-VL-7B-Instruct visual representations without fine-tuning the generative model. Mean-pooled ViT and post-merger features support only weak relation decoding for facing versus facing-away, parallel versus perpendicular, and a four-way orientation task. Linear and MLP probes generally remain near majority baselines; preserving patch-level structure yields a suggestive point estimate for parallel/perpendicular (patch-vote 73.5% vs. 64.7% majority), but that test set is small (n=34) and the confidence interval overlaps the majority baseline. In short: orientation is not robustly decodable by the tested frozen-feature readouts.

To reduce the confound that global features might infer relations without identifying the relevant objects, we additionally extract subject/reference regions and construct grounded visual features from subject, object, difference, and elementwise-product representations. Object-box conditioning with the model's own grounding predictions and mean-pooled region features does not recover robust orientation decodability. Following standard cautions about diagnostic classifiers [4], we phrase these results as decodability under the tested readouts, not as proof that the information is absent from the network. Linear decodability of visual concepts has been probed in driving-oriented VLMs [12], and directional information in this model family has been inspected via attribution [11]; our probe analysis is not the first of either kind. It differs by targeting VSR's orientation relation family with pooled, patch-vote, and object-grounded readouts over the exact conditioning features of the generative model, and by pairing decodability with adaptation and consistency interventions rather than treating it as an end in itself. Detailed probe tables and plots are in the supplementary.

Table 1. Canonical VSR test accuracy (%). Orientation improves little with model size or adaptation compared with several other relation families.

Condition	Overall	Orientation	Depth	Horizontal	Containment	Topology/contact
2B zero-shot	74.0	62.8	68.9	70.2	83.4	80.5
2B structured	68.3	53.3	64.9	66.7	78.7	71.7
2B General LoRA	76.6	62.0	71.1	74.3	87.6	81.0
2B Targeted LoRA	76.5	64.2	70.8	75.1	87.6	81.4
7B zero-shot	80.9	63.5	75.2	84.8	89.3	80.3
7B General LoRA	84.7	65.7	82.3	87.4	92.9	84.4
7B Targeted LoRA	83.9	64.2	81.7	87.1	91.7	85.3
7B Hard-Neg LoRA	84.3	66.4	80.7	87.1	89.9	84.6
7B Projector LoRA	82.9	64.2	78.6	87.1	89.3	85.3
7B Vision+Projector LoRA	83.1	64.2	78.0	87.1	89.9	84.4

5.2. Visual-Side Adaptation and Object-Centric Decomposition

If the frozen representation were the sole limiting factor, directly adapting visual or projector components might recover orientation performance. For the trained checkpoints evaluated here, it did not: projector LoRA and vision+projector LoRA both produced 64.2% orientation accuracy, below the 65.7% LM-only control, while overall accuracy was nominally lower (82.9% and 83.1% vs. 84.7%; unadjusted p=0.0043/0.0122; these effects do not survive a blanket multiple-comparison adjustment and are not load-bearing, App. D). This suggests that the tested low-rank adaptation of the visual pathway does not unlock the ceiling.

We also test a box-based two-stage decomposition. Its four-way relation module classifies facing/facing-away/parallel/perpendicular against ground-truth relation labels at 44–46% cross-validation accuracy (49.9% majority baseline; uniform four-class chance 25%), so the tested relation classifier does not outperform the empirical majority baseline. Consistent with that, the pipeline's True/False verdicts agree with the end-to-end control on only 55.1–58.3% of the 127 boxed orientation statements; in the committed analysis (10 no-box statements scored wrong), verdicts diverge on 61–65 statements, and the control is correct while the pipeline is not in 44–47 of them. The committed pipeline scored agreement with the control rather than ground-truth accuracy, so we report the tested decomposition as failing to reproduce the generative model's behavior, not as a direct accuracy comparison. Two-dimensional box geometry is structurally insufficient for intrinsic facing direction, and the tested region representations do not supply enough signal to close the gap.

5.3. Logical Consistency Reveals a Distinct Failure Mode

We construct complementary statements for strict relation families. A coherent model should return opposite truth values for left/right, front/behind, and facing/facing-away formulations of the same object pair. Complementary-pair and transformation-based consistency evaluation has prior literature, including consistency-oriented training [8]; our contribution is the VSR-specific diagnostic below, built on strict complementary relation pairs with observed verdict-pattern and base-rate analysis. The zero-shot 7B model is consistent on only 58.0% of left/right pairs, 57.0% of front/behind pairs, and 36.9% of facing pairs. The observed verdict patterns show what drives this: the model affirms either member of a facing pair only 17–20% of the time, a False-defaulting bias that yields both-False verdicts on 63.1% of pairs and no both-True verdicts (0/103); expected consistency under verdict independence given these base rates is 30.2%, below the observed 36.9% (verdict table in the supplementary): the dominant phenomenon is a refusal to affirm either proposition, rather than a preference for contradictory affirmations.

LM-only LoRA raises facing consistency to 66.0% and front/behind consistency to 70.7%; pooled strict-family consistency is significantly better than zero-shot ($p < 0.0001$). Hard-negative training further raises facing consistency to 77.7%. Facing accuracy, however, is identical for LM-only and hard-negative LoRA at 68.9%. The additional hard-negative improvement is not significant in the pooled strict-family McNemar comparison ($p = 0.29$). Paired hard-negative supervision therefore provides a targeted intervention on complementary-relation coherence: it descriptively raises facing-pair consistency while leaving original facing accuracy unchanged, although the additional

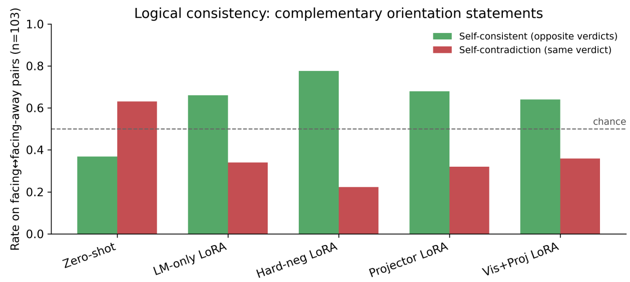

Figure 3. Self-consistency on complementary facing/facing-away relations across 7B conditions.

pooled consistency effect over LM-only is not significant. We do not claim a general coherence algorithm; the key result is separability: identical facing accuracy coexists with substantially different facing consistency. Training also changes the failure mode rather than eliminating it: both-False facing verdicts drop from 63.1% to 9.7% (LM-only) while both-True contradictions rise from 0% to 24.3%, a disclosed specialization trade-off.

Table 2. SITE cross-dataset transfer (images). Entries are raw accuracy / CAA (%).

Subset	Zero-shot	VSR-LoRA	Δ
All images	54.2 / 31.1	53.8 / 30.5	−0.4
Official spatial-rel.	75.1 / 59.2	71.6 / 53.4	−3.5
Orient. heuristic	48.3 / 24.0	48.7 / 24.5	+0.4

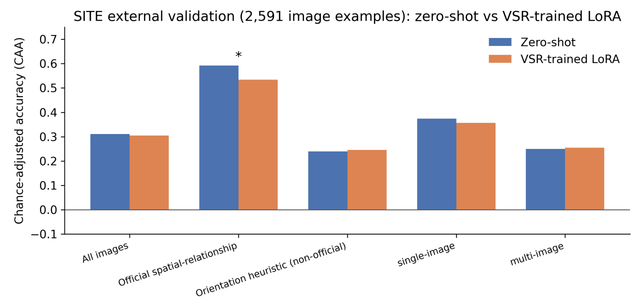

Figure 4. SITE CAA for zero-shot and VSR-trained LoRA. The official spatial-relationship category degrades significantly under transfer.

6. Cross-Dataset Transfer to SITE

We evaluate the frozen 7B base model and the VSR-trained 7B General LoRA adapter on the same 2,591 SITE image examples using SITE's native multiple-choice protocol [14], with no SITE training. Because the pre-specified decision rule resolved to branch 2 (the official category was not broadly weak), this adapter run is a post-protocol cross-dataset transfer check rather than a pre-specified step; the protocol and its outcome are documented in the supplementary. The model is strong on the official spatial-relationship-reasoning category: 75.1% raw accuracy and 59.2% CAA. Applying the VSR-trained adapter does not transfer cleanly: it is statistically neutral overall (54.2%→53.8%, $p = 0.66$) and significantly degrades the official category (75.1%→71.6%, $p = 0.004$; 52 examples fixed vs. 87 broken). VSR gains therefore transfer poorly and can induce negative transfer on the independent benchmark; we do not equate benchmark-specific fine-tuning gains with general spatial-reasoning improvement.

The pre-specified, non-official orientation-vocabulary slice is substantially weaker in the aggregate (48.3% raw / 24.0% CAA vs. 75.1% / 59.2% for the official category, and below the 54.2% / 31.1% all-images average) and shows no positive transfer under the adapter (48.3%→48.7%, $p = 0.70$). Because the slice is keyword-derived and heterogeneous (43% of its examples come from multi-view/cross-image reasoning, only 24% from the spatial-relationship category), we treat it as exploratory evidence rather than as direct replication of the VSR construct. A composition-confound analysis of the 2,591 zero-shot predictions (supplementary §E) supports this caution: after controlling for official category, source dataset, modality, and number of options, the orientation flag is not significantly associated with correctness (odds ratio 0.84, 95% CI $[0.60, 1.16]$, $p = 0.29$), although it remains descriptively 7.5pp lower within the official spatial-relationship category itself (71.3% vs. 78.8%, n=488/505). A post-hoc high-precision subset restricted to VSR-construct terms (facing, facing away, parallel, perpendicular; n=177) yields 44.1% raw (95% CI $[37.0, 51.4]$; CAA −2.3%) and is reported as exploratory only. We therefore do not treat SITE as independent confirmation of the VSR orientation bottleneck; the transfer results stand on the official category, where the model is strong and the adapter demonstrably does harm.

7. Discussion

Orientation is not spatial reasoning broadly. The strongest conclusion is deliberately narrow. The 7B model is strong on several VSR families and SITE's official spatial-relationship category. The persistent weakness concerns the object-relative orientation family in VSR; on SITE, the orientation-vocabulary slice is analyzed as exploratory evidence rather than as independent confirmation. This framing is consistent with the original VSR observation that orientation relations are difficult [7], while extending it to modern generative VLMs and a much broader intervention analysis.

Decodability and coherence both matter, as diagnostics. The probes show that orientation is not robustly decodable under the tested frozen-feature readouts, while consistency experiments expose large logical failures even for relations answered above chance. Neither result alone identifies a complete causal mechanism, and both are diagnostic rather than causal-localization claims. Together they support a mixed account in which orientation information is difficult to access reliably under the tested readouts and relational decisions are not always propagated coherently across complementary formulations.

Task adaptation can specialize rather than generalize. General LoRA improves VSR overall accuracy substantially, but its SITE transfer is neutral overall and significantly negative on the official spatial-reasoning subset. Hard negatives improve a targeted coherence behavior while producing relation-specific tradeoffs. Benchmark-specific fine-tuning gains should therefore not be equated automatically with general spatial-reasoning improvement.

8. Limitations

Our main diagnostic test contains only 137 orientation examples, and individual relation subsets such as perpendicular are smaller still (n=12); perpendicular results are therefore reported descriptively throughout. The single-annotator clean-label audit reduces ambiguity but also decreases sample size, and it is exploratory qualitative evidence rather than a ruling on label noise. The SITE orientation slice is keyword-derived rather than an official benchmark category; a composition-confound analysis shows its aggregate score is largely explained by task composition, so we do not treat SITE as independent confirmation of the VSR orientation bottleneck. Our probing methods test accessible decodability under specific linear/MLP and pooling choices; failure to decode does not prove information is absent. Visual-side experiments use LoRA rather than full end-to-end retraining. We study two differently sized VLMs from different families; this is a cross-model comparison, not a controlled scale experiment. All trained conditions are single-checkpoint runs: the paired McNemar tests quantify disagreement on the fixed test examples, not training-run variability, and these single-seed results do not estimate it. The SITE evaluation is image-only; SITE video reasoning remains future work. Finally, several interventions are evaluated as negative results, so lack of improvement should be interpreted for the tested configurations rather than as impossibility claims about all prompting, adaptation, or architectural methods.

9. Conclusion

Aggregate spatial accuracy hides a persistent orientation bottleneck in the VLMs studied here. A larger VLM and standard adaptation substantially improve several spatial families while leaving object-relative orientation comparatively resistant. Auditing, probing, visual-side adaptation, and decomposition constrain simple explanations, while complementary-relation tests show that accuracy and coherence are distinct axes of spatial competence. Cross-dataset evaluation on SITE further shows that VSR-specific adaptation transfers poorly and can negatively transfer to official spatial-relationship reasoning. The results motivate spatial evaluation that reports relation-specific performance, transfer, and logical coherence rather than relying on aggregate accuracy alone.

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